Jerome Wiesemann

PhD Student in Quantum Information
Quantum Communication | Information Theory | Convex Optimization

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Education

05/25–04/29 Expected
PhD Physics (Quantum Information)
University of Waterloo, Institute for Quantum Computing — Waterloo, ON, Canada
- Security analyses for quantum key distribution protocols with practical devices
- Contributor to the software package OpenQKDSecurity for numerical QKD analysis
- Co-president of the IQC Graduate Student Association (2025-2026)

10/22–12/24
M.Sc. Physics, GPA: 4.0/4.0
Technische Universität Berlin — Berlin, Germany
- Master's thesis: "Security implications of device imperfections in quantum key distribution," in collaboration with Fraunhofer HHI

10/19–09/22
B.Sc. Physics, GPA: 3.9/4.0
Technische Universität Berlin — Berlin, Germany
- Bachelor's thesis: "Iterative computation of individual Floquet states"

Professional Experience

10/26–Present
Quantum Cryptography Consultant
evolutionQ — Waterloo, ON, Canada
- Consulting on QKD security

02/25–04/25
R&D Internship
ID Quantique — Geneva, Switzerland
- Characterization of commercial QKD systems to address implementation security

11/22–01/25
Student Researcher
Fraunhofer Heinrich Hertz Institute — Berlin, Germany
- Analyzing vulnerabilities and hacking quantum optical implementations of QKD

06/24
1 month
Research Stay
Vigo Quantum Communication Center, Universidade de Vigo — Vigo, Spain
- Research stay resulting in [P3]

Awards

11/25
Quantum Future Award Finalist
German Federal Ministry of Education and Research
Finalist (top 5 nationwide) in Germany's national competition for master's theses in quantum technologies

07/25
Physik-Studienpreis
German Physical Society
Awarded to eight students across Berlin and Brandenburg for outstanding academic achievement for the M.Sc. and B.Sc. in physics

09/19 **Bourse au Mérite**
Région Auvergne-Rhône-Alpes
Awarded for highest distinction in the French Baccalauréat

Teaching

Dates **University of Waterloo** Waterloo, Canada
- TA for PHYS 124: Modern Physics (Winter term 2026)

10/22–03/24 **Technische Universität Berlin** Berlin, Germany
- TA for Classical Mechanics (Winter semester 2022/23)
- TA for Quantum Mechanics (Summer semester 2023)
- TA for Electrodynamics (Winter semester 2023/24)

Positions of Responsibility

10/2026 Co-organizer of the QKD Security Proof Workshop at IQC Waterloo

2025–2026 Co-president of the IQC Waterloo Graduate Student Association

Technical Skills

Programming Python, C++, MATLAB, Git

Languages German (native), French (native), English (fluent, TOEFL C2)

Invited Talks

11/24 **A security proof for BB84**
Quantum Communication Symposium Germany

12/24 **Current state of security in QKD**
German Federal Printing Office (Bundesdruckerei)

2024 **Invited talk on QKD security**
Steering committee meeting of Germany's national quantum communication initiative (QuNET)

Publications

[P1] **Security framework for practical quantum key distribution with imperfect devices**
J. Wiesemann, J. Burniston, D. Tupkary, N. Lütkenhaus
arXiv preprint | September 2026
arXiv:2609.15790

[P2] **A consolidated and accessible security proof for finite-size decoy-state quantum key distribution**
J. Wiesemann, J. Krause, D. Tupkary, N. Lütkenhaus, D. Rusca, N. Walenta
Quantum, **10**, 2037 | March 2026
doi:10.22331/q-2026-03-23-2037

[P3] **Evaluation of quantum key distribution systems against injection-locking attacks**
J. Wiesemann, F. Grünenfelder, A. Blázquez, N. Walenta, D. Rusca
APL Photonics, **10**, 066112 | June 2025
doi:10.1063/5.0260685